**16** A fixed mass of gas is contained within a sealed syringe at a pressure of 100 kPa. The plunger is slowly moved until the pressure is 80 kPa. The final volume of the gas is 50 $cm^3$. The temperature of the gas does **not** change.

What is the volume of the gas before the plunger is moved?

**A** 0.25 $cm^3$ **B** 40 $cm^3$ **C** 63 $cm^3$ **D** 160 $cm^3$